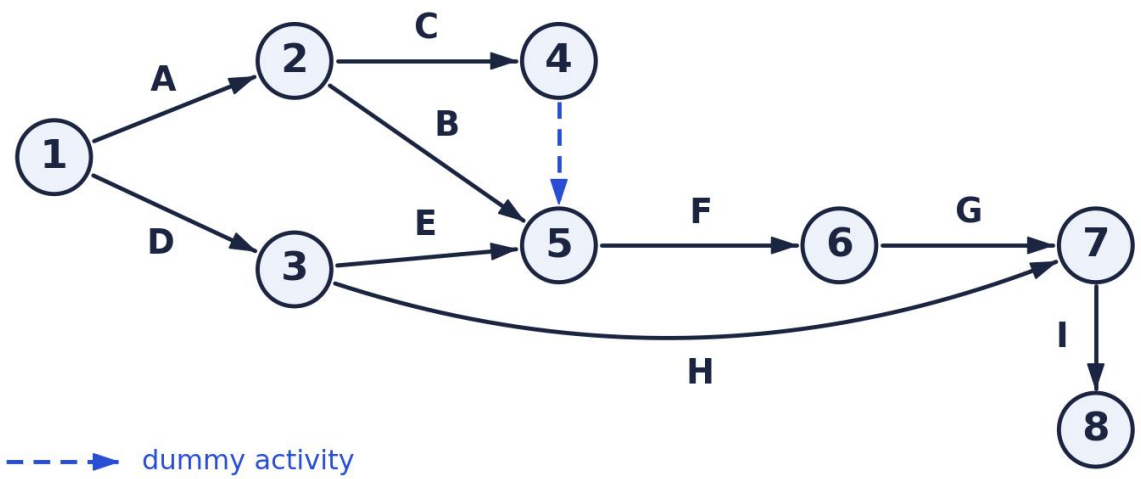
Task 1

Draw the Pert chart for the following activities:

Activity	Predecessor
A	-
B	A
C	A
D	-
E	D
F	B, C, E
G	F
H	D
I	G, H

Answer

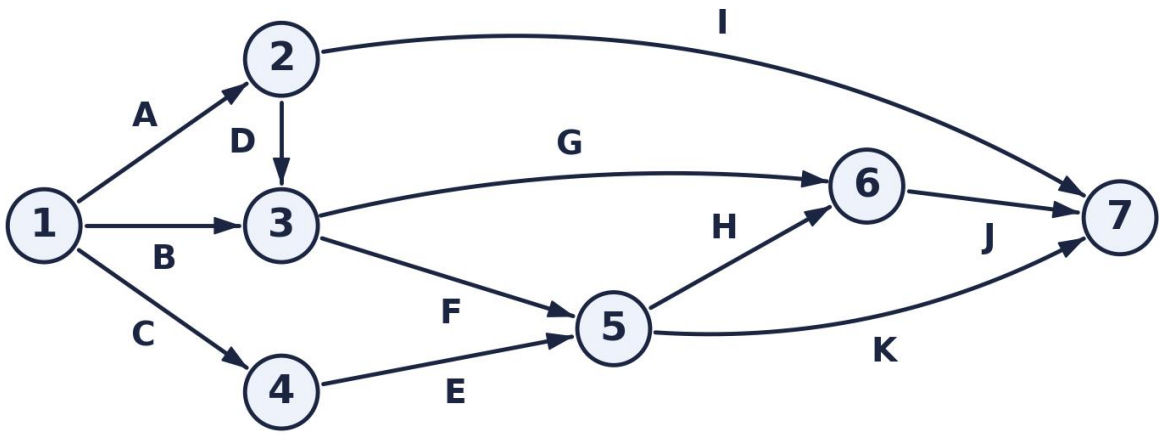


Task 2

Draw the Pert chart for the following activities:

Activity	Predecessor
A	-
B	-
C	-
D	A
E	C
F	B, D
G	B, D
H	E, F
I	A
J	G, H
K	E, F

Answer

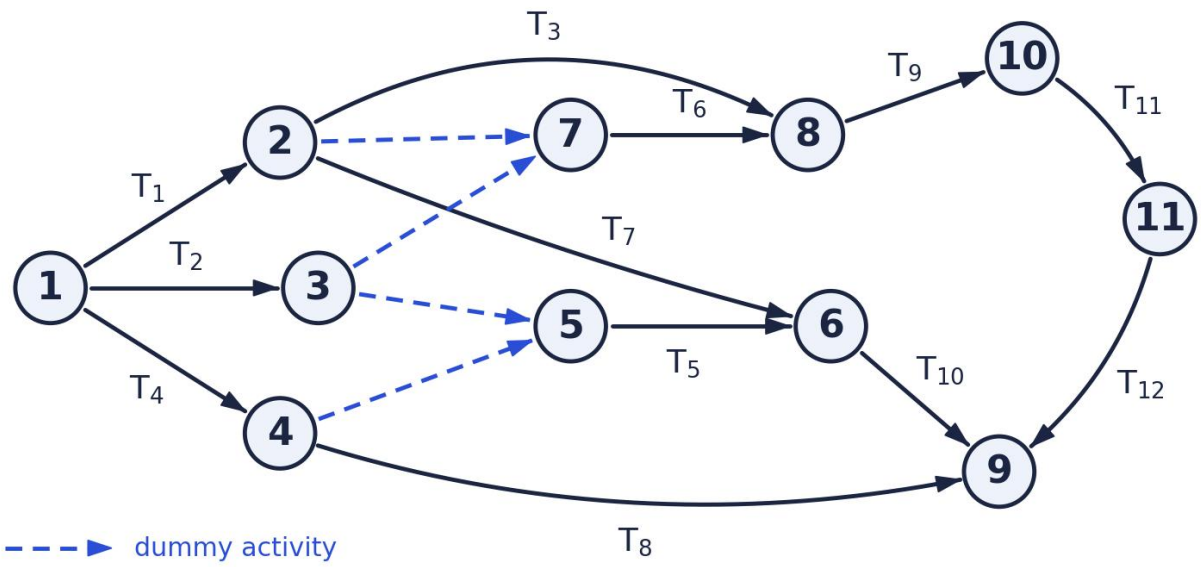


Task 3

For the following activities, draw the Pert and Gantt chart, then determine the minimum project duration and the critical path:

Activity	Duration (Days)	Predecessor
T1	8	-
T2	15	-
T3	15	T1
T4	10	-
T5	10	T2, T4
T6	5	T1, T2
T7	20	T1
T8	25	T4
T9	15	T3, T6
T10	15	T5, T7
T11	7	T9
T12	10	T11

Answer: Pert chart



Minimum project duration & critical path

Minimum project duration = 55 days

Critical path: T1 → T3 → T9 → T11 → T12

8 + 15 + 15 + 7 + 10 = 55 days. These activities have zero float (ES = LS), so any delay in one of them delays the whole project. The other activities have float (T2 = 3, T4 = 20, T5 = 15, T6 = 3, T7 = 12, T8 = 20, T10 = 12 days) and can be delayed by that amount without affecting the finish date.

Answer: Gantt chart

